

# TCA CYCLE

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# Stages of Oxidation of Foodstuffs

## □ First Stage

Digestion in the gastrointestinal tract converts the macromolecules into small units. This is called **primary metabolism**.

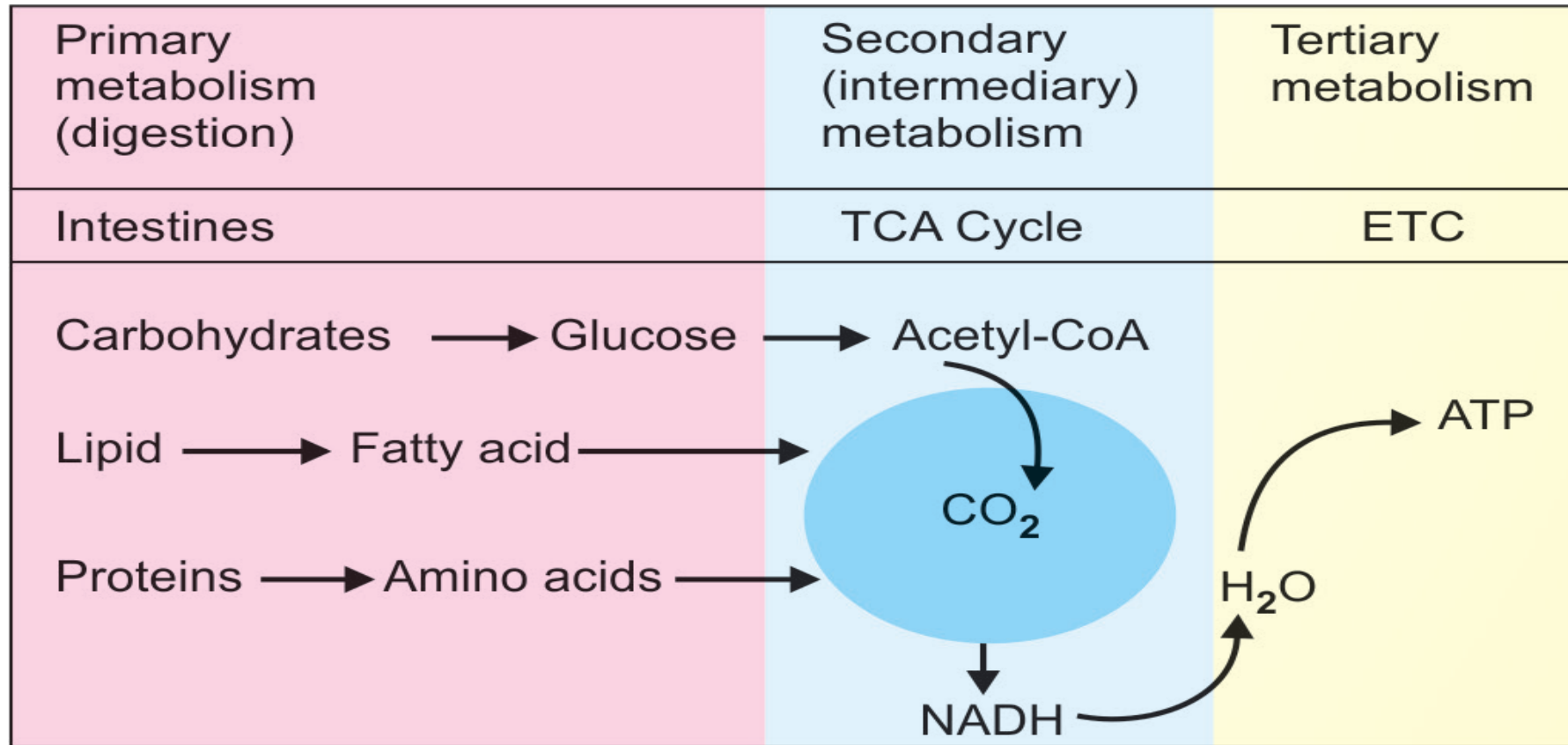
## □ Second Stage

The products of digestion are absorbed, catabolized to smaller components, and ultimately oxidized to CO<sub>2</sub> in the citric acid cycle. In this process, NADH and FADH<sub>2</sub> are generated. This is called **secondary or intermediary metabolism**.

## □ Third Stage

These reduced equivalents (NADH and FADH<sub>2</sub>) enter into the **electron transport chain** (ETC), or **Respiratory** chain, where energy is released. This is the **tertiary metabolism** or **internal respiration** or cellular respiration

# Oxidation of Foodstuffs in Three Stages



The TCA cycle oxidizes acetyl-coenzyme A (acetyl-CoA) derived from carbohydrates, fatty acids, amino acids, and ketone bodies, and produces NADH and  $\text{FADH}_2$  for the electron transport chain

# Citric Acid Cycle

- Also called Krebs cycle or tricarboxylic acid-TCA cycle.
- Most important metabolic pathway for energy supply to the body.

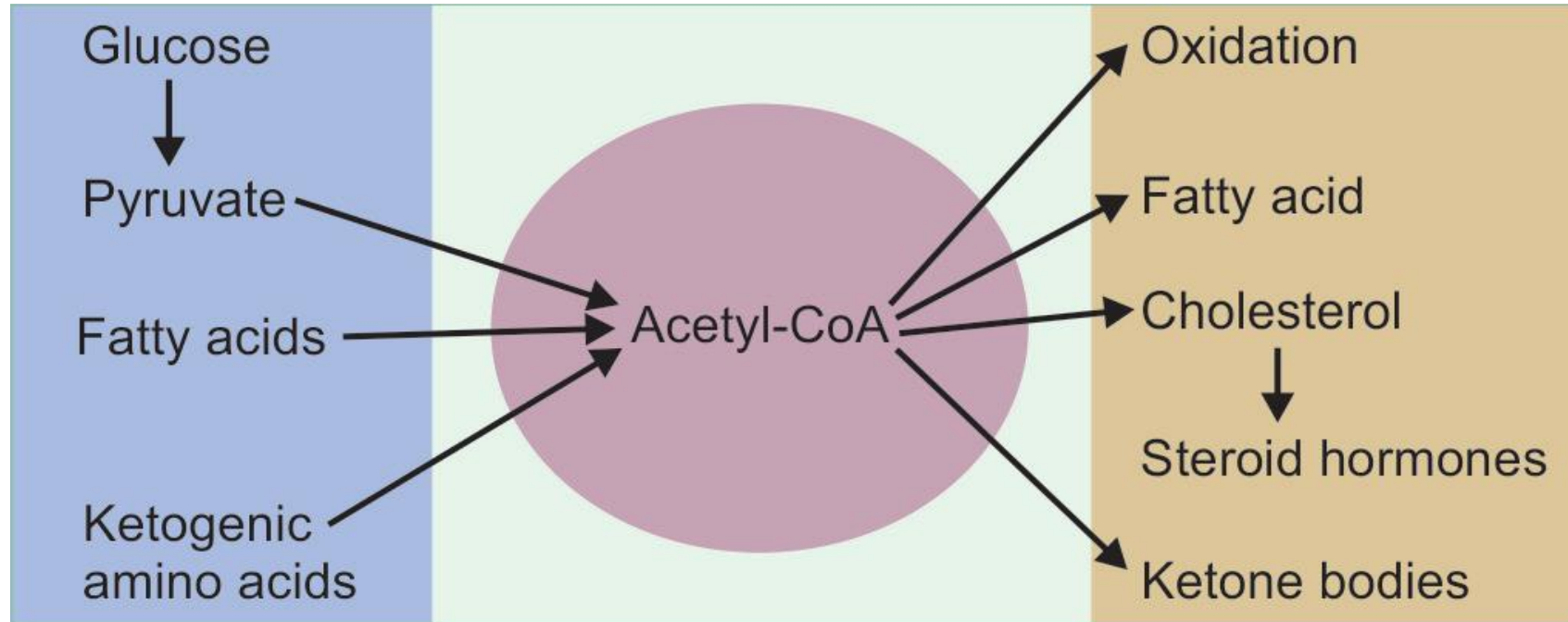
# Functions of the Citric Acid Cycle

1. Source of reduced co-enzymes that provide the substrate for the respiratory chain.
2. It acts as a link between catabolic and anabolic pathways (amphibolic role).
3. Provides precursors for synthesis of amino acids and nucleotides.
4. Final common oxidative pathway that oxidises acetyl CoA to CO<sub>2</sub>.

# Location of TCA cycle

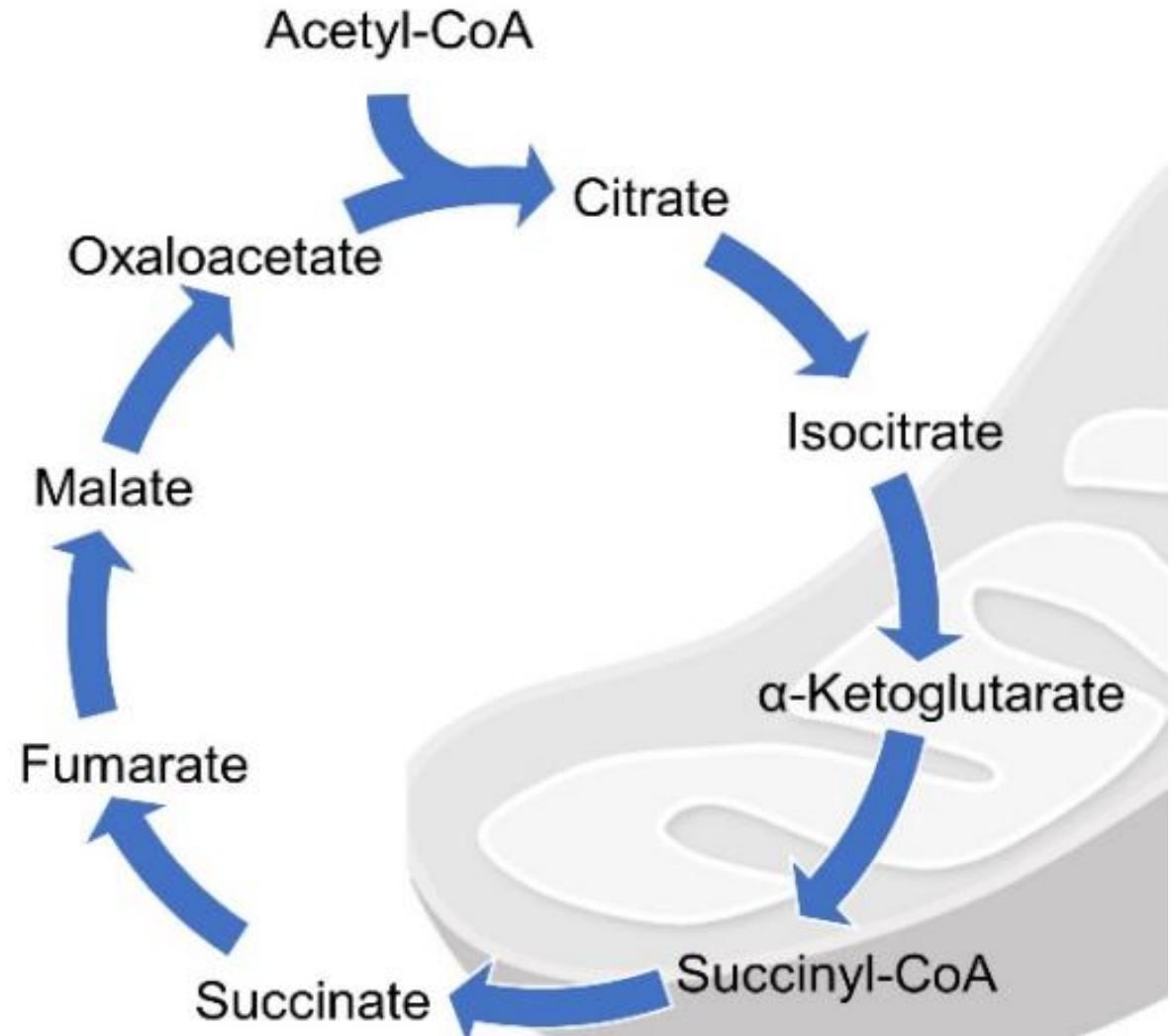
- The TCA cycle, occurring in the **mitochondria**, generates large amounts of energy under aerobic conditions

# Sources and utilization of acetyl-CoA.



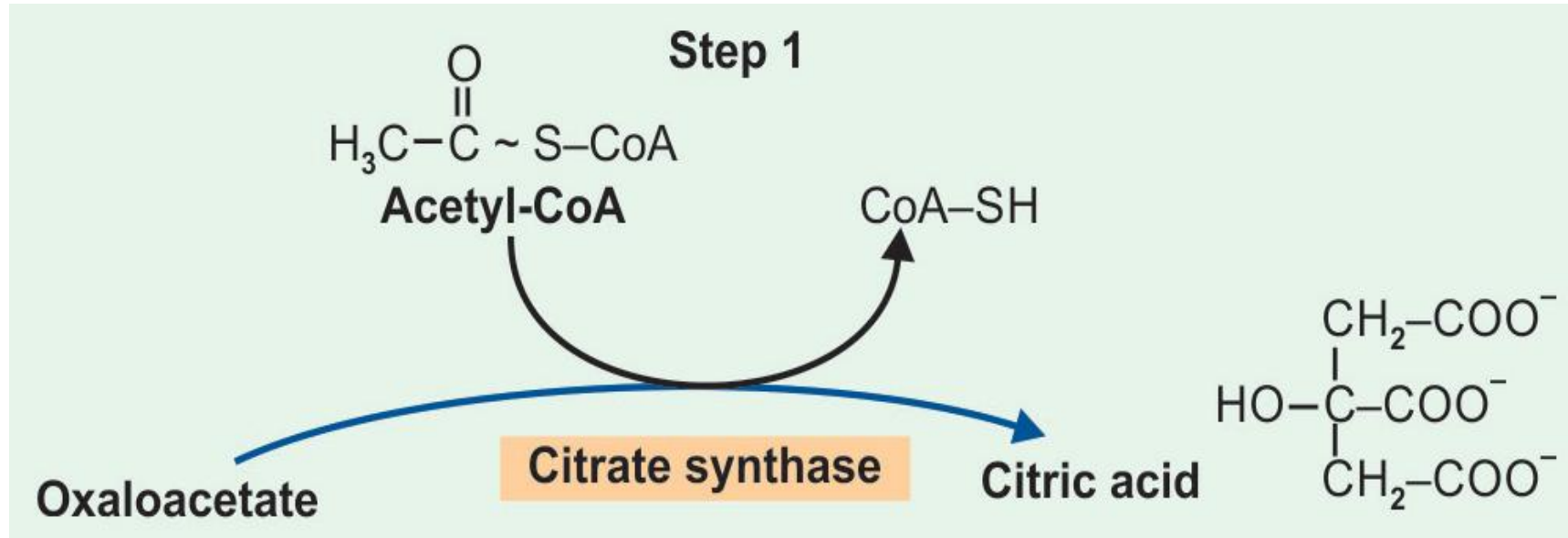
# Citric Acid Cycle

- Krebs cycle is cyclic process, many compounds enter the cycle and leave.
- It is comparable to a heavy traffic circle in a national highway with many connecting roads. Each intermediate of the cycle connecting another pathway is a road.

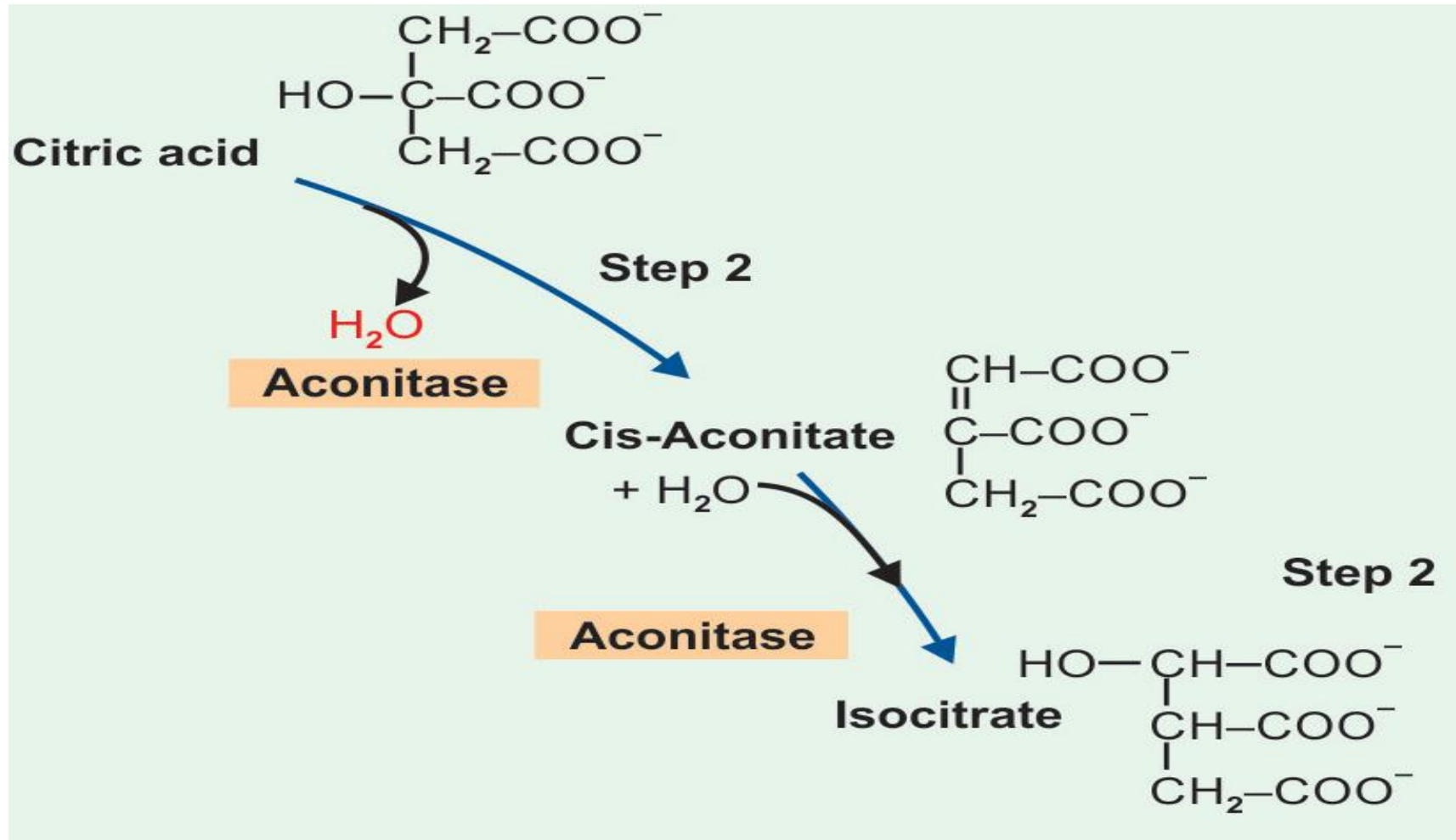




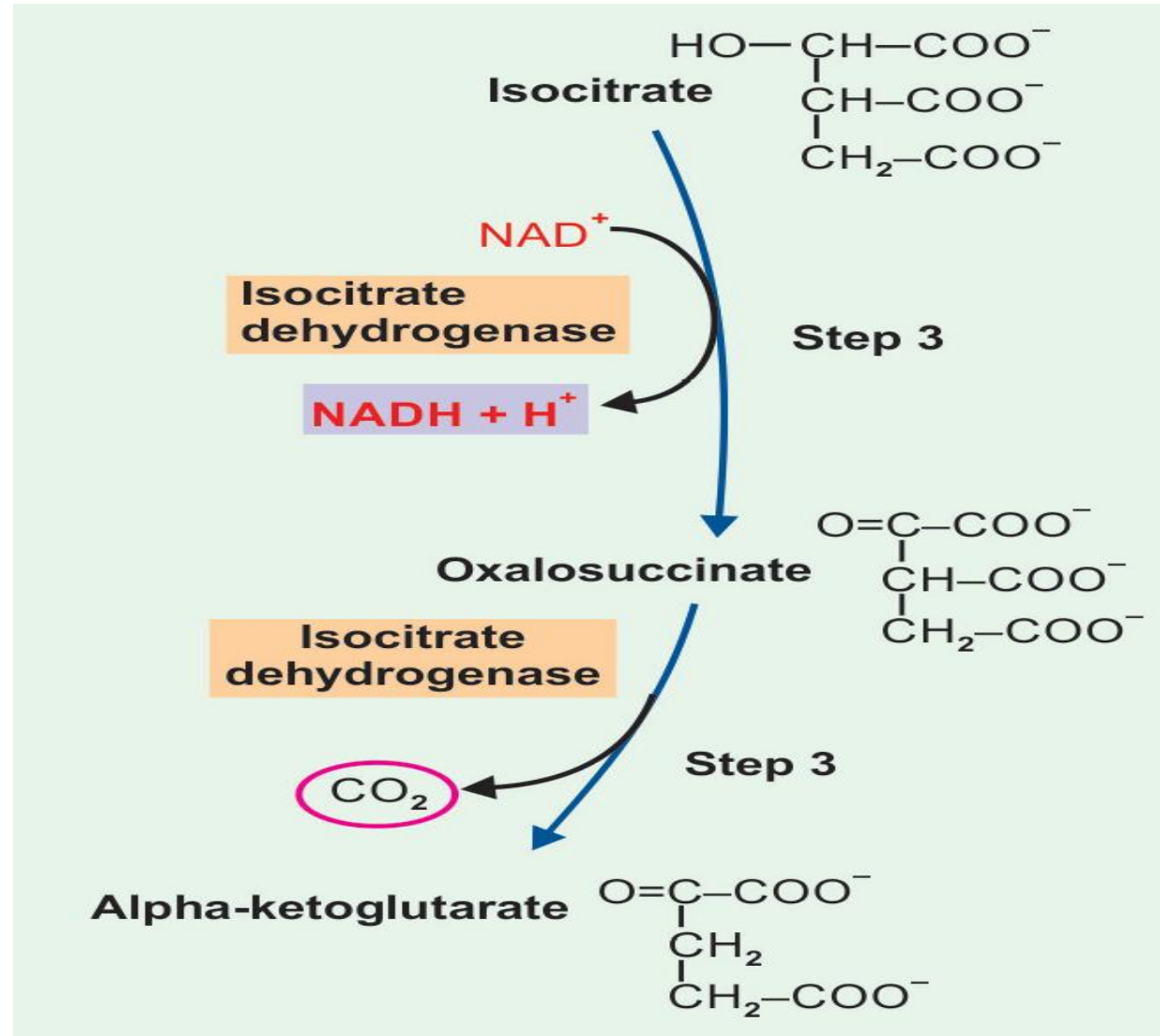
# Step 1 of citric acid cycle.



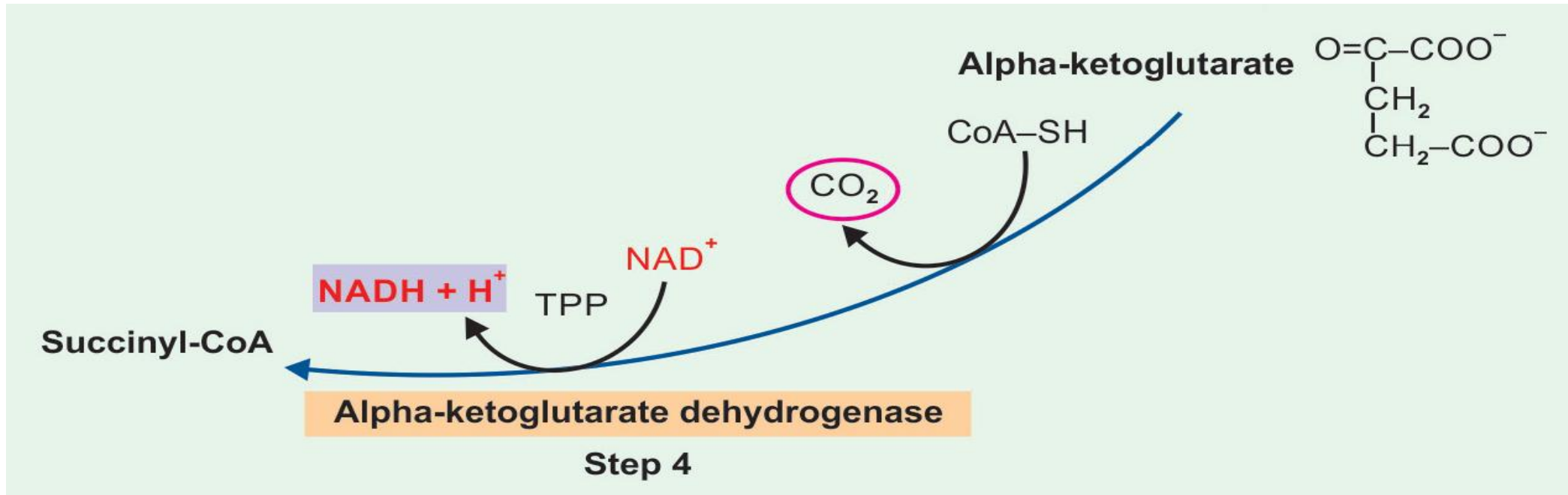
# Step 2 of citric acid cycle.



# Step 3 of citric acid cycle.

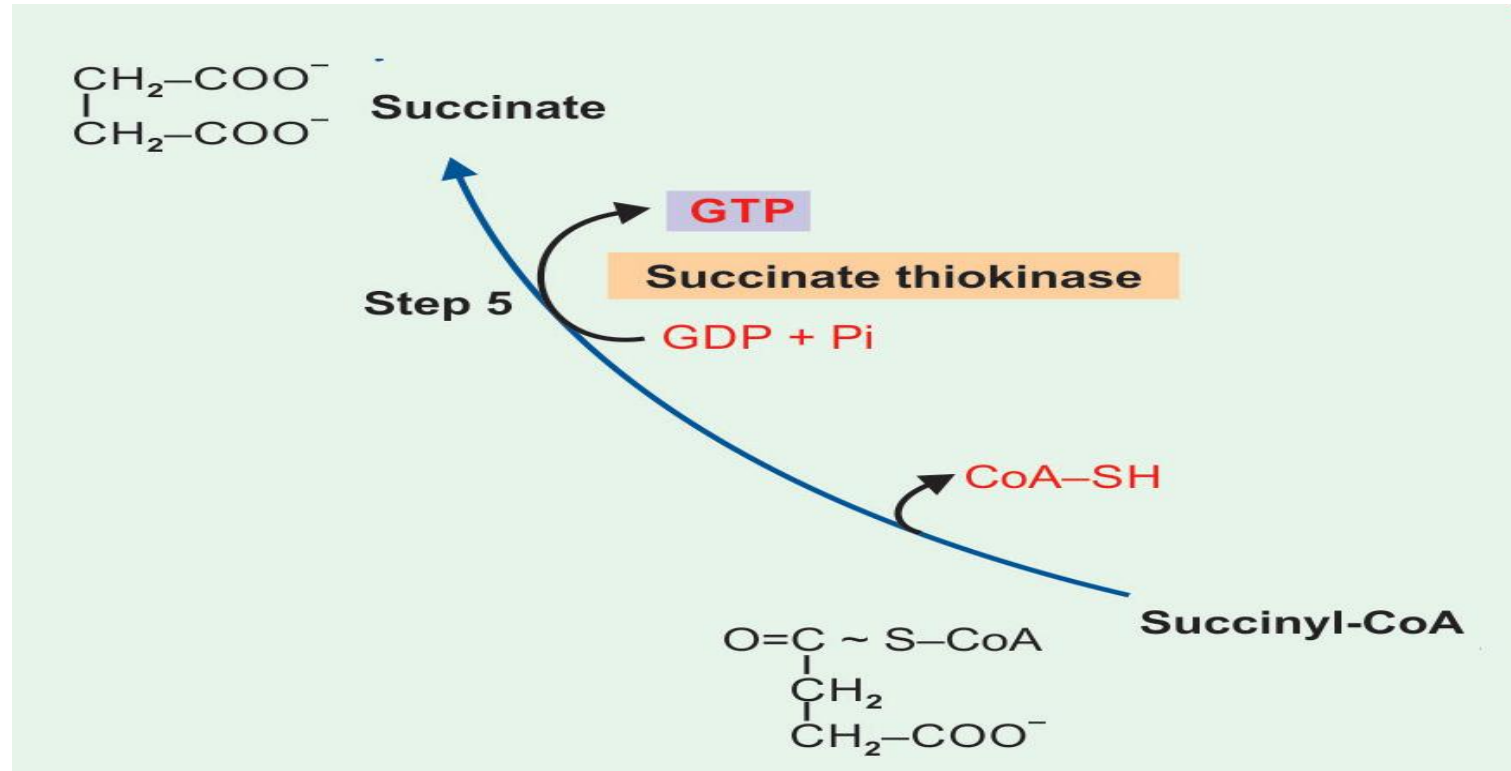


# Step 4 of citric acid cycle.



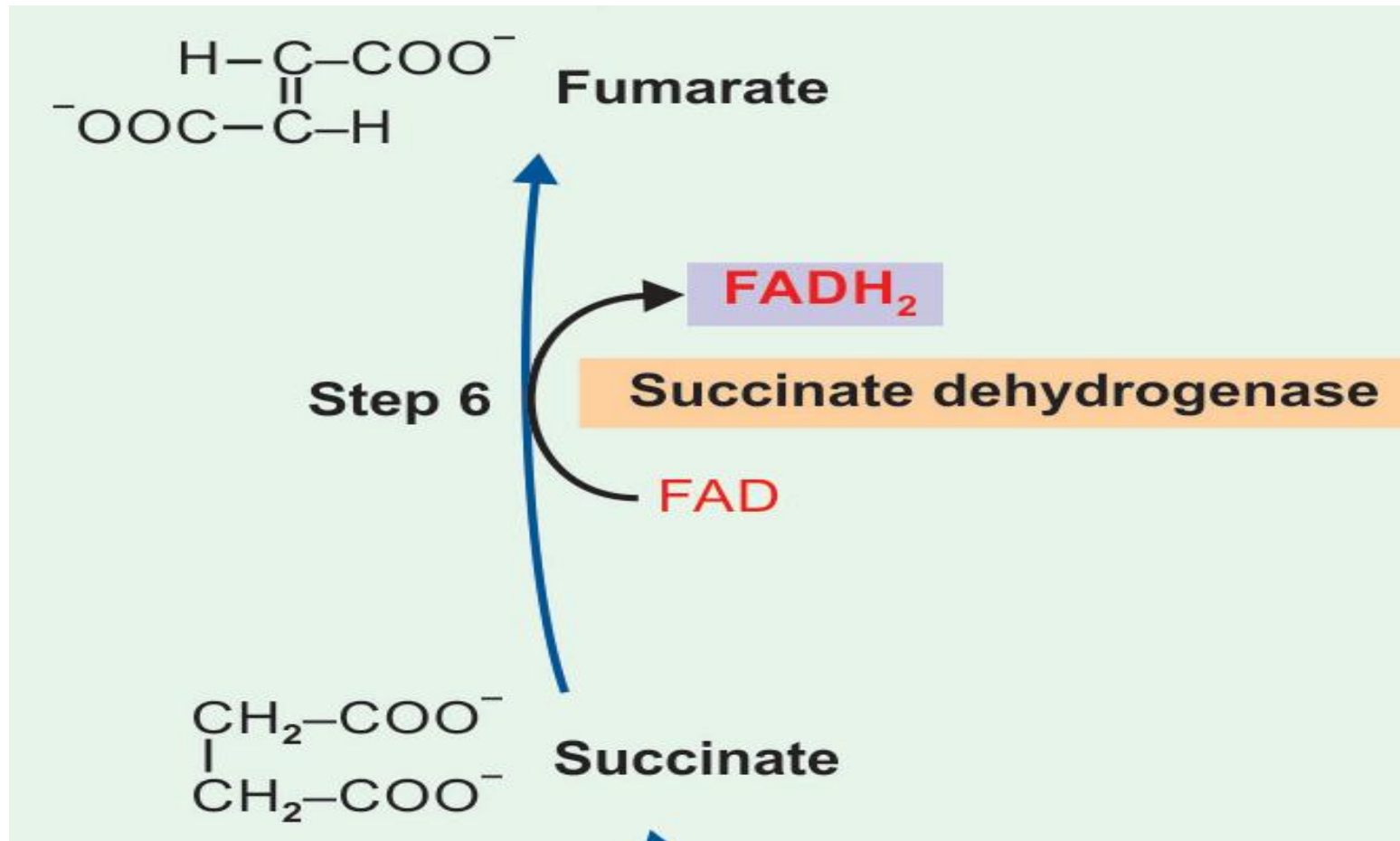
- $\alpha$  - Ketoglutarate Dehydrogenase is multienzyme complex.  
(similar to PDH Complex)
- 3 enzymes:  $\alpha$  - Ketoglutarate Dehydrogenase  
Trans succinylase  
Dihydrolipoamide dehydrogenase
- Using 5 co-enzymes like Thiamine pyrophosphate(TPP),  
Lipoamide, NAD<sup>+</sup> , FAD, CoA

# Step 5 of citric acid cycle.

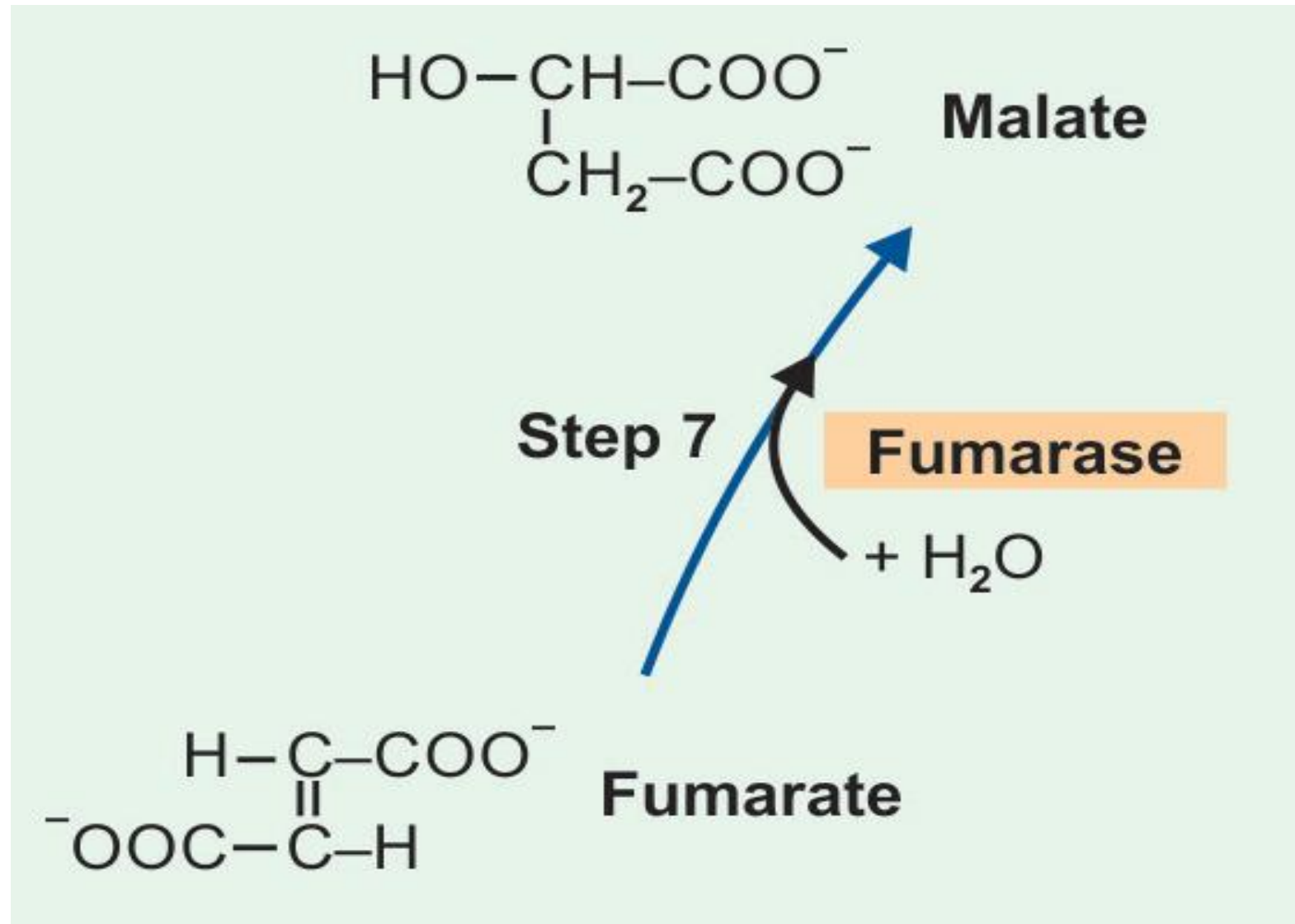


- Substrate level phosphorylation step
- Reaction upto formation of succinate are referred as “**decarboxylation phase**” and remaining reactions as the “reductive phase” to regenerate oxaloacetate.

# Step 6 of citric acid cycle.

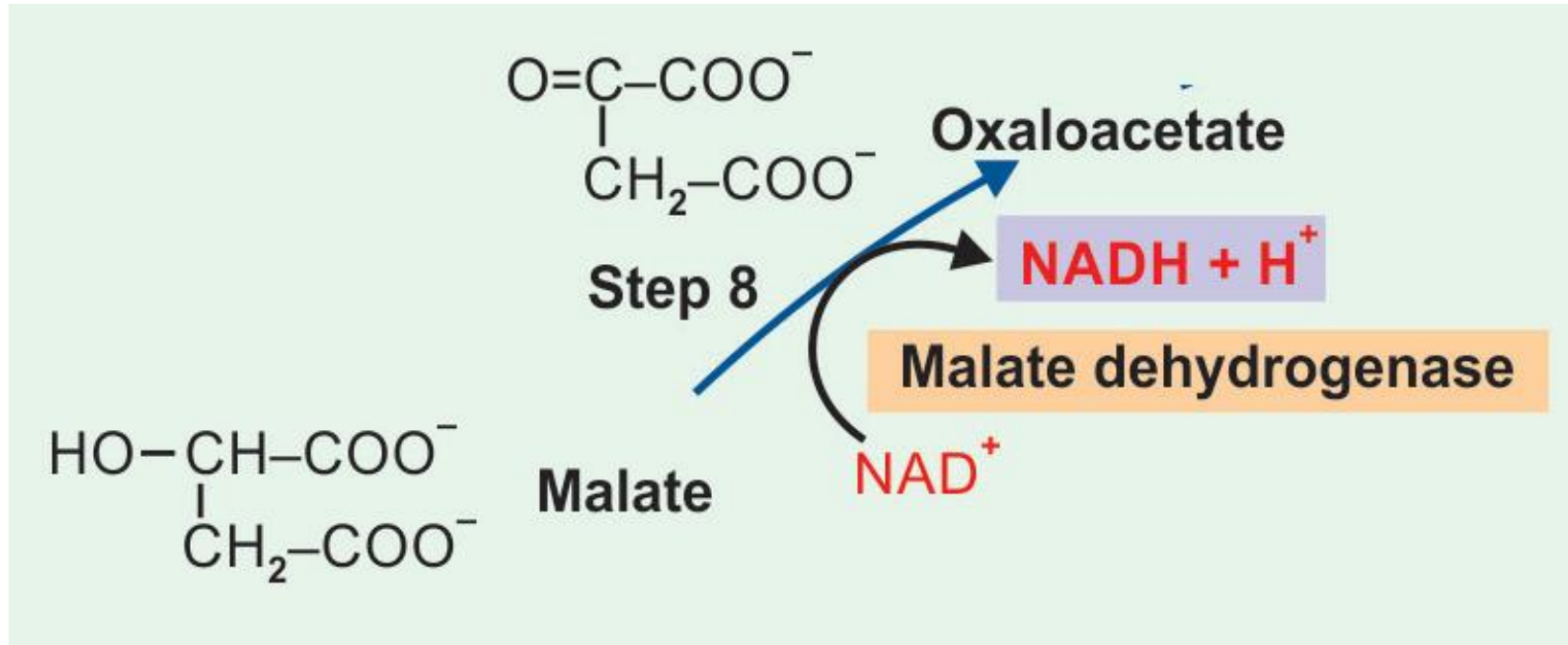


## Step 7 of citric acid cycle.

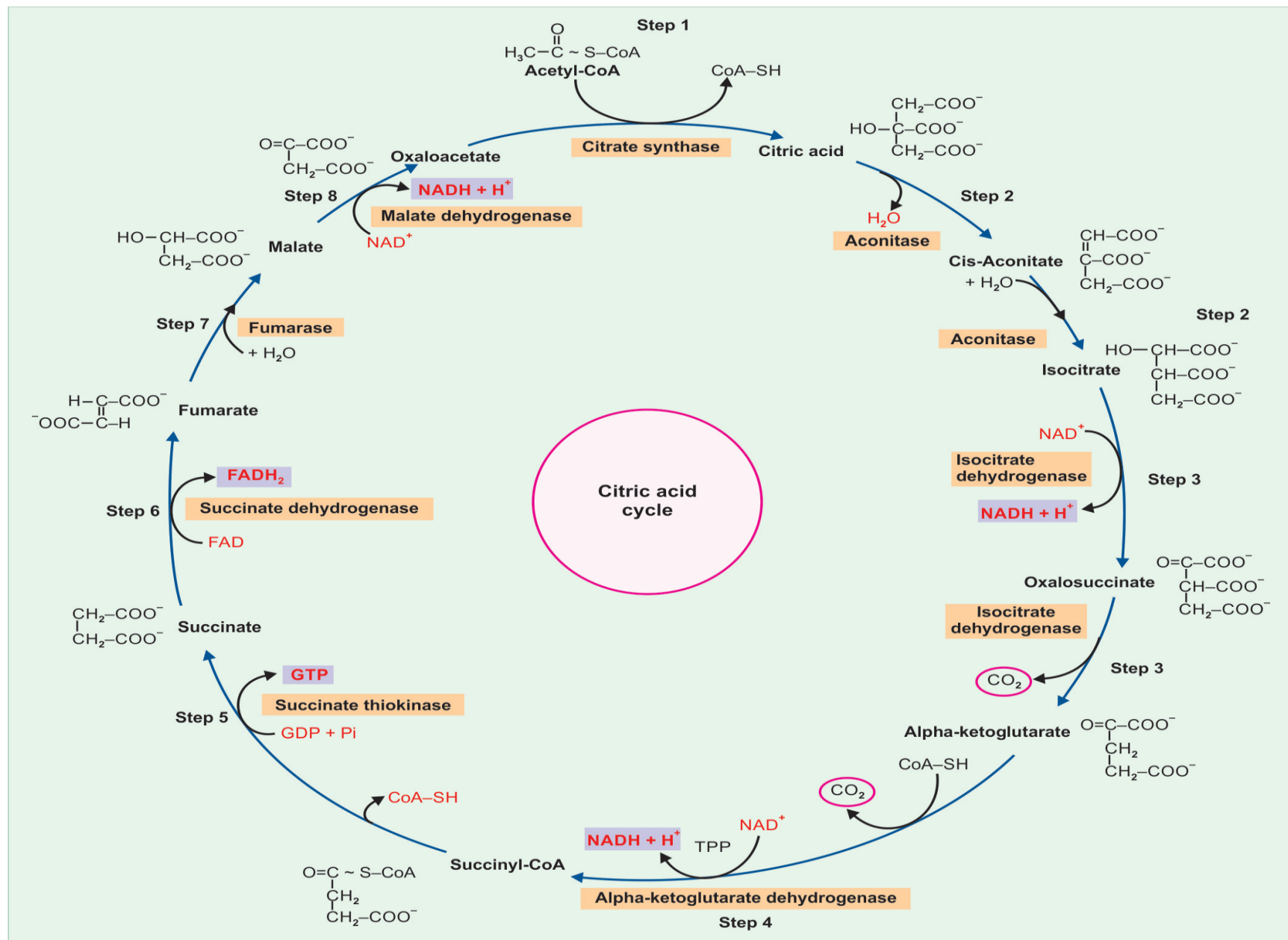




## Step 8 of citric acid cycle.



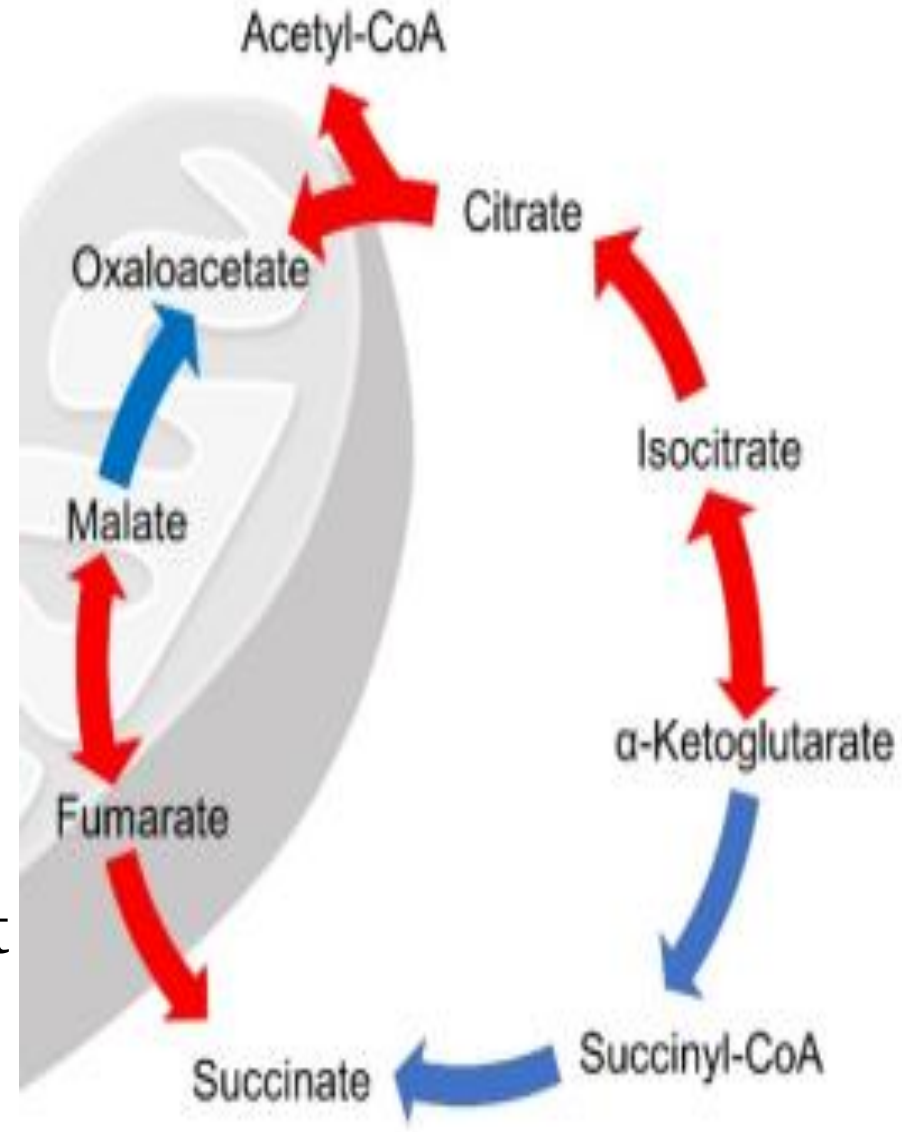
- The oxaloacetate further condense with another acetyl co-A molecule and the cycle continues.



# ATP Generation Steps of Citric Acid Cycle

| Step No. | Reactions no.                                  | Co-enzyme         | ATPs generated |
|----------|------------------------------------------------|-------------------|----------------|
| 3        | <u>Isocitrate</u> → <u>alpha ketoglutarate</u> | NADH              | 2.5            |
| 4        | Alpha ketoglutarate → succinyl CoA             | NADH              | 2.5            |
| 5        | Succinyl CoA → Succinate                       | GTP               | 1              |
| 6        | Succinate → Fumarate                           | FADH <sub>2</sub> | 1.5            |
| 8        | Malate → Oxaloacetate                          | NADH              | 2.5            |
|          |                                                | Total             | 10             |

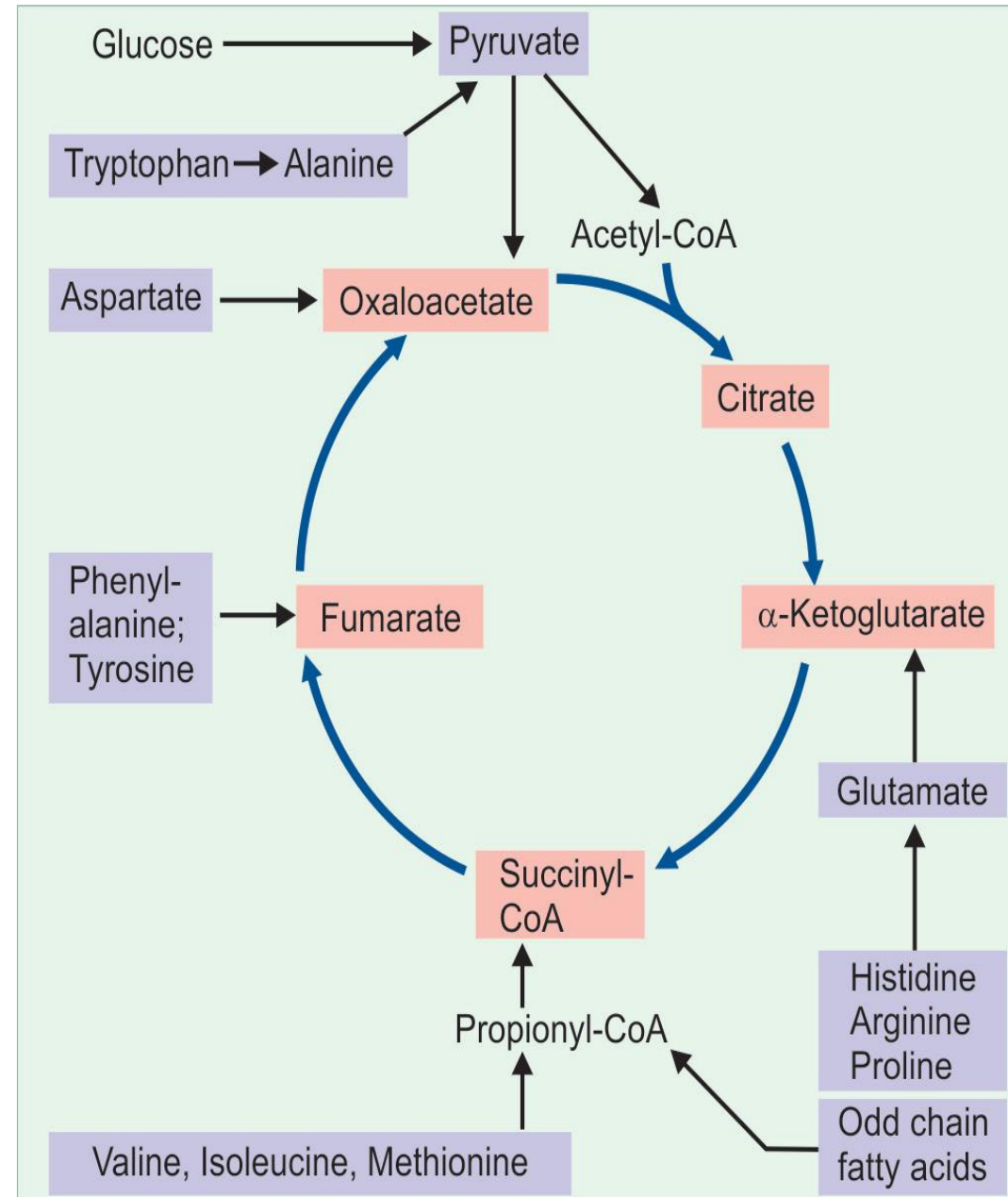
- Acetyl-CoA (2 carbon), enters the cycle.
- These carbon atoms are released as CO<sub>2</sub> in the steps 3 and 4. So acetyl-CoA is completely oxidized by the time the cycle reaches alpha-ketoglutarate.
- All reactions are reversible except the 1<sup>st</sup> and 4<sup>th</sup> steps.



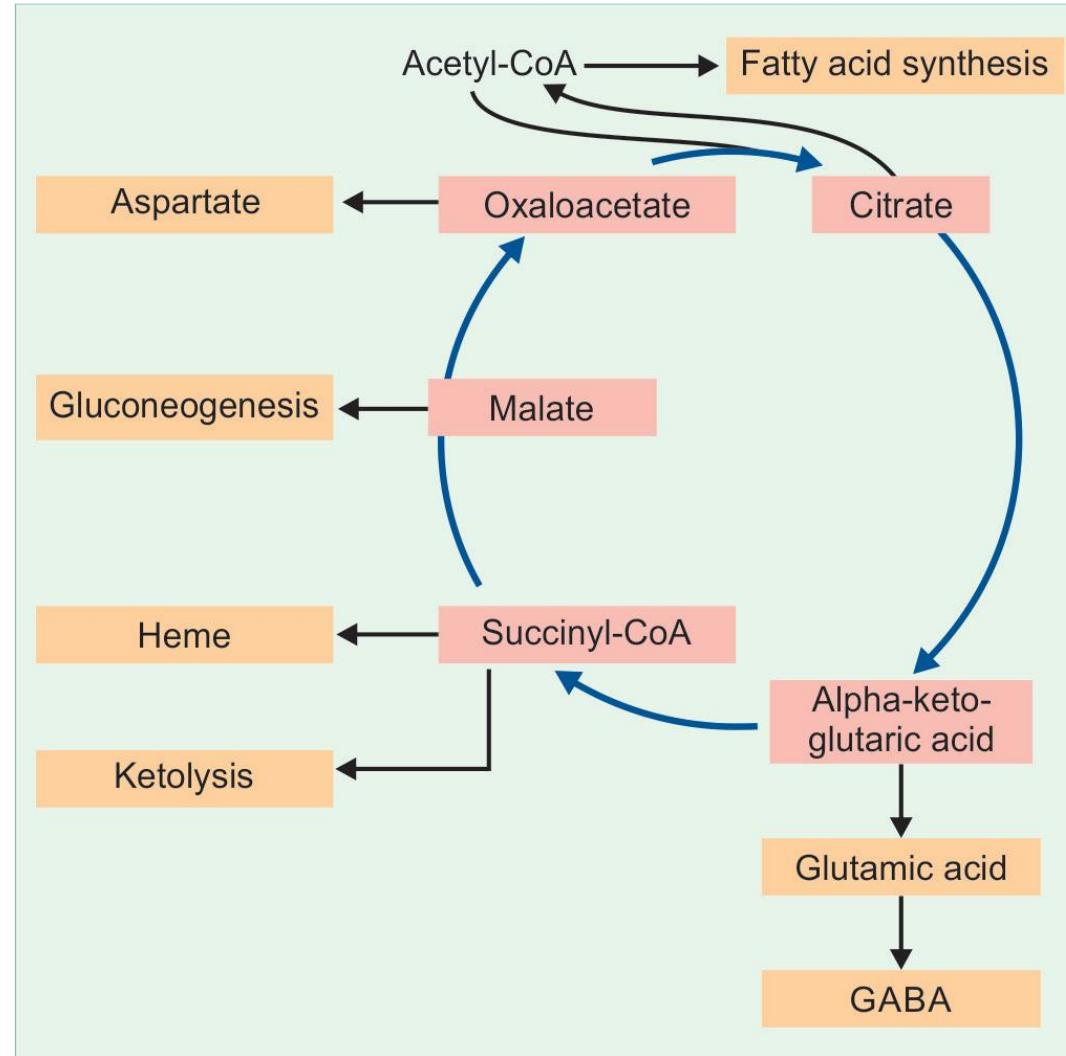
# Significance of Citric Acid Cycle

1. Complete oxidation of acetyl-CoA( $\text{CO}_2$  removal steps like 3 and 4).
2. ATP generation
3. Final common oxidative pathway of all foodstuffs.
4. Integration of major metabolic pathways
5. Carbon skeletons of amino acids finally enter the citric acid cycle
6. Amphibolic pathway
7. Anaplerotic role

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Influx of TCA cycle intermediates. Amino acids entering the cycle are shown in violet boxes



# Efflux of TCA cycle intermediates



# Regulation of TCA Cycle

1. Citrate Synthase:- Inhibited by citrate; ATP
2. Iso Citrate Dehydrogenase:- Inhibited by NADH
3.  $\alpha$ -KG Dehydrogenase:- Inhibited by succinyl Co-A and NADH.
4. Availability and cellular need for ATP



Reactions through which TCA Cycle Intermediates are Replenished from other Precursor Molecules

# Anaplerotic Reactions

| Precursor                                                                          | TCA Cycle Intermediate  | Reaction & Enzyme                                                                                                                                                                         |
|------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aspartate</b> →                                                                 | <b>Oxaloacetate</b>     | Reversible Transamination by <b>AST</b>                                                                                                                                                   |
| <b>Glutamate</b> →<br>(Which can be synthesized from Arginine, Histidine, Proline) | <b>α keto Glutarate</b> | Reversible Transamination by <b>AST / ALT</b> OR Deamination by <b>Glutamate Dehydrogenase</b>                                                                                            |
| <b>Odd Chain Fatty Acids, Branched Chain Amino Acids</b> →                         | <b>Succinyl CoA</b>     | Propionyl CoA (Synthesized from the Odd Chain Fatty Acids and branched chain Amino Acids) is converted to Succinyl CoA by <b>Propionyl CoA Carboxylase &amp; Methylmalonyl CoA Mutase</b> |
| <b>Adenylosuccinate</b> →                                                          | <b>Fumarate</b>         | Catalyzed by <b>Adenylosuccinate Lyase</b><br>Occurs in Purine Metabolism                                                                                                                 |
| <b>Phenyl Alanine, Tyrosine</b> →                                                  | <b>Fumarate</b>         | Cleavage of Phenyl Alanine catabolic intermediate Fumarylacetoacetate                                                                                                                     |

# Inhibitors of TCA Cycle

- **Fluoroacetate** : non-competitive type  
(Enzyme inhibited :Aconitase)
- **Arsenite** : non-competitive type  
(Enzyme inhibited: $\alpha$ -KG dehydrogenase)
- **Malonate** : competitive type  
(Enzyme inhibited :Succinate dehydrogenase)  
Malonate has structural similarity to succinate